

NIKHIL AGRAWAL

+91-7587685847

niknir136@gmail.com

nikhilagrawal

nikhilagrawal

Leetcode:// nikhilxgawal

Experience

Snapdeal (Acevector Group)

Software Engineer I

September 2024 – Present | Gurugram, Haryana

- **Architected a centralized, extensible AI Orchestration microservice** using **Java 17** and **Spring Boot 3+**, providing a unified abstraction layer to seamlessly integrate and swap different LLMs including **Amazon Bedrock** and **Google Vertex AI**. Implemented dynamic routing, circuit breakers, and centralized prompt management to optimize token usage and response latency for high-throughput ecommerce flows.
- **Engineered a production-grade post-order chatbot** using **Spring Boot 3** and **Vertex AI**, automating complex customer journeys for returns, refunds, and order tracking. Leveraged **Aerospike** for high-speed session persistence and **AWS S3** for interaction logging, resulting in a **3x reduction** in customer support call volume.
- **Revamped the Ads Seller & Admin panels** by developing a high-performance frontend-integrated backend and an automated reporting engine, generating **₹15L daily revenue**. Built automated data-processing jobs to deliver deep-funnel analytics to stakeholders, driving **~₹50L monthly savings** by eliminating dependency on external third-party vendors.
- **Re-engineered Ads conversion tracking pipelines** using **Kafka event processing** and **HBase persistence**. Optimized the system by decoupling workloads into specialized cron jobs, REST APIs, and Kafka consumers, while migrating the infrastructure from EC2 to **Kubernetes** to save **~\$300/month** in operational costs.
- **Enhanced the VIP Subscription engine** by implementing a rule-based personalization layer using an in-house **Java-based A/B testing framework**. This enabled real-time dynamic pricing at checkout and improved audience targeting accuracy through intent-based feature flags.
- **Integrated Amazon Bedrock** into the core search service to handle zero-result queries. Developed an intent-detection layer to rewrite failed queries into relevant search terms, directly improving the search-to-detail page conversion rate.

Graduate Engineering Trainee

January 2024 – September 2024 | Gurugram, Haryana

- **Streamlined user onboarding** by integrating the **Truecaller SDK** into the Snapdeal App and M-site authentication microservices. Reduced friction in the login/signup journey, leading to a **20% improvement** in successful onboarding rates.
- **Developed an AI-assisted vernacular address translation pipeline** using **Google Translate API** and **Vertex AI** to standardize regional Indian addresses into English. This reduced RTO (Return to Origin) rates and delivery failures caused by courier non-compliance with vernacular formats.

Projects

AI-Powered Ecommerce Platform - Spring Boot + MCP Integration | Github

December 2025

- **Developed a custom MCP (Model Context Protocol) server** from scratch using **Java 17**, **Spring Boot 3+**, and **Spring AI** to expose Snapdeal's product catalog for semantic search and discovery.
- **Integrated with Claude Desktop** as an MCP client to enable a conversational shopping experience, bridging the gap between LLM reasoning and live production data.
- **Architected for high availability** by deploying the server on **Kubernetes** within a private environment, ensuring secure and scalable natural language product discovery.

Technical Skills

Programming Languages: Java 17/8, Python

Frameworks: Spring Boot 3.x/2.x, Microservices, Hibernate, REST APIs, Apache Kafka, Spring AI

Tools: Linux, Git, Kubernetes, Jenkins, Grafana, Pinpoint, Docker

Databases: MySQL, Cassandra, HBase, Aerospike, AWS S3

AI/LLM: Google Vertex, Amazon Bedrock, MCP (Model Context Protocol)

Education

Jaypee Institute of Information Technology

2020-2024 | Noida, Sec. 62, India

Bachelor of Technology in Computer Science & Engineering

Cum. GPA: 8.1/10.0

Leadership / Awards

Snapdeal (Acevector Group)

Q1 FY 2025-26 | Gurugram, Haryana

For contributions to high-impact cross-functional projects (AdTech, Conversion Tracking, and Reporting systems).